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Machine learning-driven market value prediction for European football players

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ABSTRACT

Football is globally recognized as the most widely practiced and watched sport. Precise player value is crucial for clubs seeking to maximize their player acquisition strategy and overall success in football. Conventional player valuation methodologies are mainly dependent on expert judgments and subjective assessments, missing the objectivity and precision provided by data-driven approaches. This study seeks to close this disparity by utilizing machine learning techniques to predict the market valuations of football players. The analysis is conducted using an extensive dataset sourced from the FIFA 22 video game, which was obtained via sofifa.com. The collection includes more than 16,000 players. The Machine Learning (ML) techniques used in this study are Multiple Linear Regression (MLR), Ridge Regression (RR), Support Vector Regression (SVR), and Random Forest Regression (RFR). The machine learning algorithms undergo training using 80% of the samples and are subsequently tested using the remaining 20% of the samples. We evaluate each algorithm's performance using metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) value. Numerical results show that the RFR model demonstrates superior performance by achieving the lowest MAE, MSE, RMSE, and the highest R^2 value across all samples. The RFR effectively captures non-linear interactions and reliably prevents overfitting. This research finding will enhance the existing knowledge in sports economics by demonstrating how ML can be used to anticipate the market prices of football players with better accuracy. This will provide football teams with valuable insights to make more strategic decisions.

1. Introduction

Football is the most popular sport globally [1]. A significant transformation in contemporary football is player's and clubs' widespread adoption of professionalism. Nowadays, football has evolved into an industry that engages players on a full-time basis, with clubs investing substantial financial resources toward their players and infrastructure. The football industry has always been highly profitable, but in recent years, the financial stakes have reached unprecedented levels. European football clubs generated an estimated $27 billion in revenues in 2017, making football a substantial contributor to the global economy [2]. Football Benchmark's annual evaluation of Europe's 32 leading football teams based on Enterprise Value (EV) predicts that in 2024, the 9th edition of the "Football Clubs' Valuation: The European Elite" report emphasizes the increasing importance of football as a top-tier genre of

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entertainment. The total EV of the top 32 clubs has reached EUR 59.1 billion [3]. This implies a surge of 124% compared to the initial edition in 2016 and a growth of 14% from the preceding year [3]. In 2019, Real Madrid Club de Fútbol (Real Madrid CF), a Spanish football club, earned a total of EUR 757 million. By 2024, the club's revenue had risen and totaled EUR 831 million [4]. The 27th edition of the Deloitte Football Money League reports that the overall revenue of football teams hit EUR 9.2 billion in 2023 and rose to EUR 10.5 billion in 2024, representing a 14% increase compared to the previous year [4]. In recent decades, there has been an enormous rise in the market for great football players, resulting in player prices surpassing EUR 200 million. This value is considerably higher than historical trade amounts and the average inflation rate [5]. By 2024–2025, players from around the world will have begun competing at the most elite levels of the sport, and top clubs will frequently travel across continents to take part in global competitions. Football clubs must make a crucial decision when selecting players as it has a big impact on their likelihood of achieving success [6–8]. Player transfers have a significant impact on a club, causing experts from several fields to research the factors that influence transfer fees [5,9,10].

In recent times, there has been a growing interest in the market valuations of football players. The market value of a player reflects the potential amount that a team could obtain by selling the player's contract to another team [11]. Transfer fees precisely reflect the specific amounts paid in transactions, whereas market values provide essential estimations that are important for transfer agreements. Football specialists, including team management and sports media, have consistently emphasized the significance of market valuations. In recent years, crowdsourcing platforms such as Transfermarkt.com have emerged as trustworthy sources for determining market values [12]. The website includes comprehensive football-related information, including scores, results, news, transfer rumors, and market value estimates for both individual players and teams in various professional football leagues. Video games like Electronic Arts Fédération Internationale de Football Association (FIFA) and eFootball Pro Evolution Soccer (eFootball PES) serve as useful data sources for football analytics, specifically about FIFA. Football clubs have been using these games as alternative data stores since 2014.

Machine learning methods have drawn a great deal of attention in recent years in regression and other statistical analyses [13–17]. An excellent review of these methods is available in [18,19]. A machine learning model has been developed by employing the FIFA 20 dataset and implementing of Particle Swarm Optimization (PSO) method with a combination of Support Vector Regression (SVR) method. The model underwent training using 55 different features. The results demonstrated that this approach could accurately assess the value of players with an accuracy level of 74% [2]. Different research was analyzed by employing Linear Regression (LR), Multiple Linear Regression (MLR), and Random Forest Regression (RFR) [20] algorithms. Standard accuracy measures [21] such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) were employed here. This analysis was based on 7 distinct features, resulting in R^2 values of 0.47 for LR, 0.67 for MLR, and 0.95 for RFR [22]. In addition, a model was created using Multi-level Regression with 22 characteristics, which produced an MAE of 3,241,733 [5].

In other sectors as well, in recent years, ML techniques have gained a lot of interest [23–28]. A machine learning-based inverse method integrating finite element analysis, optimization, and data-driven techniques (e.g., neural networks) efficiently identifies constitutive model parameters, achieving accurate results 2–3 times faster than traditional optimization across diverse material models [29–31]. Boosting machine learning models (XGBoost, LGBoost, AdaBoost) combined with nonlocal strain gradient theory effectively predict thermal vibration frequencies in functionally graded porous nanobeams, with XGBoost achieving near-perfect accuracy ($R^2 = 0.999$) and Shapley Additive exPlanations (SHAP) analysis revealing foundation parameters as most influential [32–35].

The study aims to identify the best attributes for football players based on their positions through detailed analysis and apply machine learning algorithms to predict player market values with improved accuracy. The objectives include developing a robust machine learning model for precise market value prediction, comparing algorithms like MLR, RR, SVR, and RFR, and showcasing the practical application of machine learning in sports economics, specifically in football player valuation.

2. Related work

Numerous studies have employed machine learning models for calculating the market value of football players. A demonstrative model, utilizing the FIFA 20 dataset, implemented a fusion of PSO with SVR. This approach achieved a commendable accuracy of 74% in estimating player values [2]. A Multi-level Regression model was constructed utilizing data from 4217 players across six seasons in the top five European leagues [5]. This study identified age, position, country, player performance, and popularity as significant determinants of market worth. Moreover, it emphasized the constraints of the crowd-sourced estimation method employed by transfermarkt.com. A further study conducted on 150 prominent attackers utilizing the Generalized Least Squares (GLS) technique revealed that goals, assists, and team value had a substantial influence on market worth. Additionally, it was proposed that FIFA ratings could play a significant role. However, concentrating exclusively on a certain player type may not yield all-encompassing observations [6]. A mechanism was established to assess the market value of 12,858 players using performance data. The approach revealed enhanced accuracy in predicting team performance when making use of transfermarkt.com market value estimations [36]. A different research study introduced a technique for calculating transfer fees based on factors such as age, accuracy, achievement, confidence, and adaptability. The study found that age had a negative impact on the market value. Finally, a study was conducted to examine the influence of talent and popularity on market value. This inquiry employed 20 criteria to assess talent. The study utilized an Ordinary Least Squares (OLS) regression model to determine that player popularity has a strong positive impact on market value [37]. An alternative method utilized data from 2259 players participating in the top five football leagues in Europe (Spain, England, Italy, France, Germany) during the 2017/2018 season [38]. This model examined the relationships among factors such as age, position, player caliber, and team caliber. The findings demonstrated how variables such as team level, birth month, league, playing position, and age had a major effect on market value. The data used for this analysis was obtained from transfermarkt.com.

Table 1
Dataset Details of EA Sports 2022.

Dataset	Total data (Sample size)	After cleaning & feature extraction	Forward	Mid-Fielder	Defender	Goalkeeper
EA Sports FIFA 2022	16 710	12 915	1990	5512	4279	1134

3. Dataset description

This study utilizes the EA Sports FIFA video game data, specifically obtained from *sofifa.com*, a well-regarded digital football platform. Transfermrkt.com offers extensive data and statistics about football, including player biographies, league details, and market worth evaluations. The software has an extensive database that allows several forms of football analytics and market analysis. It provides significant insights into player performance and valuation. The EA Sports FIFA video game series, developed and distributed by Electronic Arts, has become globally recognized as an incredibly popular and dependable reservoir of football information since its establishment. The game simulates the experience of real-life soccer, allowing users to influence teams and players from many international leagues. Since 2014, researchers and football clubs have frequently used the data from EA Sports FIFA game due to its outstanding level of accuracy and deep nature [28,39–45].

This study uses the EA Sports FIFA 22 dataset, encompassing information from 16,710 individuals. Each player is characterized by more than 70 distinctive characteristics. These attributes provide an in-depth evaluation of player performance across multiple dimensions, encompassing physical features, technical skills, and mental attributes. Moreover, the dataset offers significant insights into team dynamics, player potential, contract details, and estimations of market value. Physical qualities include specific attributes such as age, height, and weight. Technical talents include abilities like dribbling, ball handling, crossing, passing, shooting, and volleys. Mental qualities include various factors like visual acuity, emotional stability, strategic positioning, quick reflexes, endurance, and the ability to anticipate and disrupt opponents' activities. The physical and movement attributes encompass pace, acceleration, sprint speed, jumping ability, and balance. On the other hand, the defensive attributes consist of standing tackle, sliding tackle, and marking. The dataset also includes goalkeeping traits such as diving, kicking, handling, placement, and reflexes. The comprehensive structure of the FIFA 22 dataset enables a thorough examination of player qualities, yielding significant insights into the elements that influence a player's overall performance and market worth [46]. The study intends to utilize the vast amount of data available to construct prediction models using machine learning techniques. These models attempt to estimate player valuations more accurately and objectively compared to conventional methods. The dataset offers comprehensive profiles for every participant, covering their physical characteristics, technical abilities, and mental traits. This level of detail provides a thorough analysis of player strengths and potential futures. The player ratings are consistently revised based on real-life performance, ensuring that the data remains pertinent and precise. This feature is essential for the analysis of future events and the assessment of a player's worth. The dataset contains comprehensive financial information, including market value, wages, and contract period. This data provides a strong foundation for economic analysis and studies on transfer markets. Accessibility to data on teams and leagues allows for the analysis and comparison of performance across various contests. Player potential and development trajectories provide valuable information about future great players and can help with long-term team strategy. The ratings are an indicator of the fluctuations in player performance that occur during several seasons, effectively representing the continually evolving nature of player development and decline. This element is crucial for trend analysis and forecasting [41–45]. The comprehensive player qualities and the dynamic rating system establish a strong basis for examining many facets of football performance and economics.

4. Methodology

The methodology of this study involves a systematic approach to analyzing the FIFA 2022 dataset to predict football player performance and market values using various machine learning models. The process is outlined in the accompanying structural outline, which illustrates the key steps and stages involved in the analysis (Fig. 1).

4.1. Data collection

Table 1 provides detailed information about the EA Sports FIFA 2022 dataset, which initially includes data from 16,710 players. After the processes of cleaning and feature extraction, the dataset size is reduced to 12,915 players. The table categorizes the players into various positions: forward (1990 players), midfielder (5512 players), defender (4,279 players), and goalkeeper (1134 players). This categorization helps in understanding the distribution of players across different playing positions following the data cleaning and feature extraction steps.

4.2. Data cleaning and attribute selection

This involves managing null values, converting categorical variables into numerical representations, and normalizing numerical characteristics to make the dataset consistent. Football players are classified into four main positions: goalkeepers, defenders, midfielders, and forwards. Every position necessitates specific traits that have a substantial impact on the players' ratings and performance. The chosen features of a goalkeeper (GK) include Age, Potential, GK Diving, GK Handling, GK Kicking, GK Positioning,

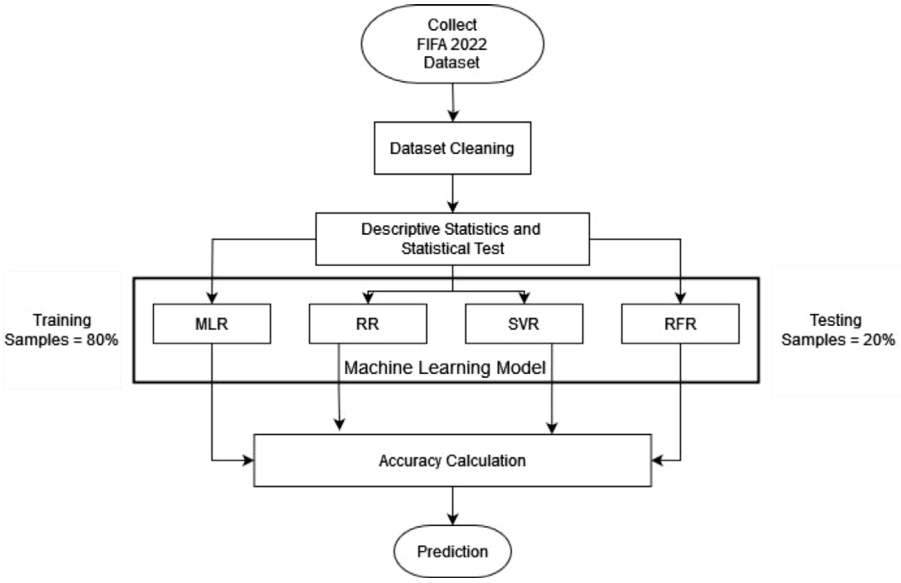


Fig. 1. Structural outline of the entire study.

Table 2
Features and target variables for each player category.

Player category	Features	Target variable
Goalkeeper	Age, Potential, GK Diving, GK Handling, GK Kicking, GK Positioning, and GK Reflexes	Market Value
Defender	Age, Potential, Height, Passing, Balance, Strength, Interceptions, Sliding Tackle, Work Rate, and Minimum BMI	
Midfielder	Age, Potential, Crossing, Short Passing, Dribbling, Long Passing, Ball Control, Sprint Speed, Long Shots, Aggression, Vision, Work Rate, and BMI	
Forward	Age, Potential, Finishing, Heading Accuracy, Short Passing, Volleys, Dribbling, Ball Control, Sprint Speed, Shot Power, Penalties, Work Rate, and BMI	

and GK Reflexes. The chosen features of a defender include Age, Potential, Height, Passing, Balance, Strength, Interceptions, Sliding Tackle, Work Rate, and Minimum BMI. The chosen features of a Midfielder include Age, Potential, Crossing, Short Passing, Dribbling, Long Passing, Ball Control, Sprint Speed, Long Shots, Aggression, Vision, Work Rate, and BMI. The chosen features of a Forward include Age, Potential, Finishing, Heading Accuracy, Short Passing, Volleys, Dribbling, Ball Control, Sprint Speed, Shot Power, Penalties, Work Rate, and BMI. The selected target variable for all the categories of players is Value, which indicates the market value of players(see Table 2).

4.3. Descriptive statistics and statistical tests

An analysis of the dataset involves the use of descriptive statistics and statistical tests to examine the distribution and relationships between different variables. The data is condensed by computing fundamental statistical measures such as the average and variability, which offer insights into the core tendency and spread of the data. The mean is a statistical measure that determines the average value for each attribute, serving as a central point of reference. However, the standard deviation measures the amount of variation or spread from the average, indicating the level of consistency or variability in player qualities.

Furthermore, accurate statistical tests are employed. The Jarque–Bera (JB) test [47] is applied to evaluate the normality of a distribution, the Durbin–Watson (DW) test [48,49] is utilized to detect autocorrelation, and the t-test is used to compare means between different groups. These studies provide useful insights into the key factors that influence player performance and market valuations, enabling a deeper understanding of the dataset's characteristics and connections.

Table 3
Hyperparameters used in the Models.

Models	Hyperparameters
RR	regularization parameter $\lambda = 1.0$
SVR	Kernel = 'rbf', Gamma (Kernel Coefficient) = 'scale', C (Regularization Parameter) = 1.0, Epsilon (ε-insensitive Tube) = 0.1
RFR	n_estimators = 100, random_state = 42

4.4. Machine learning models

A supervised learning [50,51] methodology is employed to construct prediction models for evaluating the market value of football players. This methodology allows the models to acquire knowledge of the fundamental patterns and connections within the data, hence enabling precise forecasts on novel, unobserved occurrences. This study implements four supervised learning algorithms: MLR, RR, RFR, and SVR. The algorithms are utilized to categorize football players into four distinct groups based on their playing positions: goalkeepers, defenders, midfielders, and strikers. The models can incorporate the distinctive characteristics and performance measurements that are pertinent to each position by classifying the players. 80% of the data is allocated as a training sample, while the remaining 20% is allocated as a testing sample.

$$\boldsymbol{LR}: y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Where $\beta_1, \beta_2, \beta_3, \ldots \beta_n$: Coefficients of the independent variables, β_0: Intercept (bias term), $x_1, x_2, x_{3,} \ldots, x_{n,}$: Independent variables (features) and y: Dependent variable (target).

$$\boldsymbol{RR}: \hat{\beta} = \min_{\beta} \left(\sum_{i=1}^{n} (y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right)$$

Where $\hat{\beta}$ = The estimated coefficients for the regression model, β_0 = The intercept term, y_i = The observed value of the dependent variable for the ith observation, x_{ij} = The value of the jth predictor for the ith observation, β_j = The regression coefficient associated with jth predictor and λ = The regularization parameter.

$$\boldsymbol{SVR}: f(x) = \omega^T \varnothing(x) + b$$

Where ω^T: Weight vector of the model, $\varnothing(x)$: Feature mapping function (possibly non-linear) that transforms x into a higher-dimensional space and b: Bias term.

$$\boldsymbol{RFR}: f(x) = \frac{1}{M} \sum_{m=1}^{M} f_m(x)$$

Where M: Number of trees in the random forest and $f_m(x)$: Prediction of the mth tree.

Also, the Variance Inflation Factor (VIF) is selected as a means of identifying potential multicollinearity in the analysis. The statistical metric in question is employed in the MLR model to evaluate the extent of multicollinearity among the predictor variables. The presence of multicollinearity is responsible for the observed rise in variance of the estimated regression coefficients with VIF.

4.5. Model accuracy metrics

The models are evaluated based on evaluation metrics: MAE, MSE, RMSE, and R^2. These methods evaluate the models to help compare different models, select the best-performing model, and identify areas for improvement in model performance [52–55]. The evaluation helps in determining the accuracy and efficiency of each model.

$$MAE = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_i)^2}$$

Where n = Total observations in the dataset, y_i = The actual values and $\hat{y}_i$ = The predicted values.

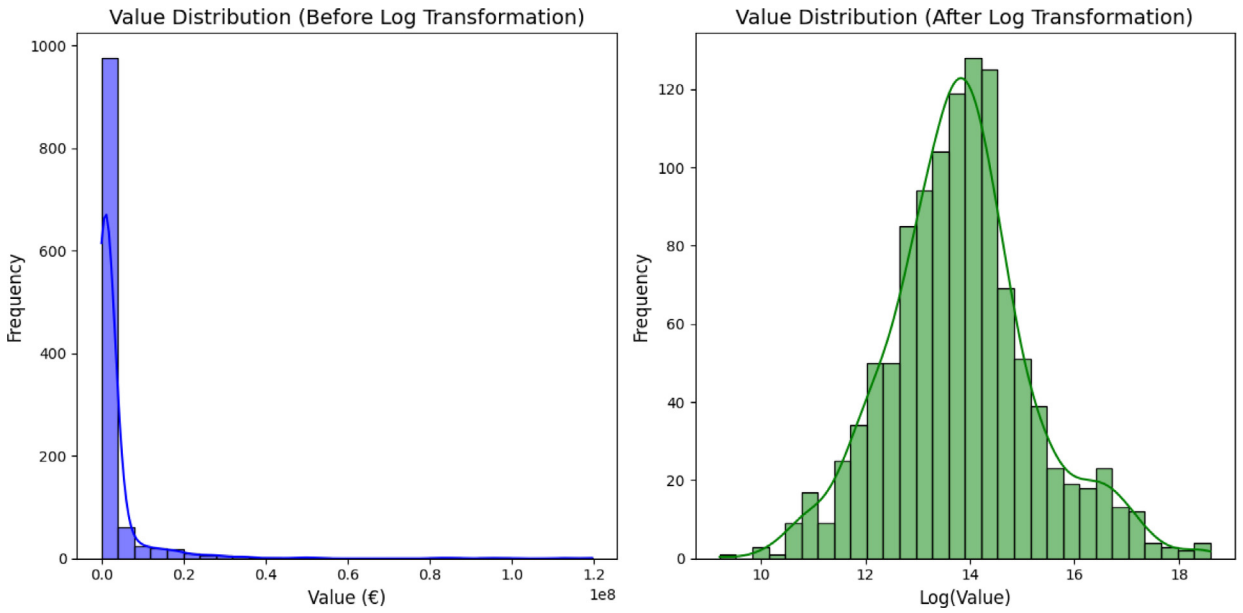


Fig. 2. Symmetric distribution of market value of forwards.

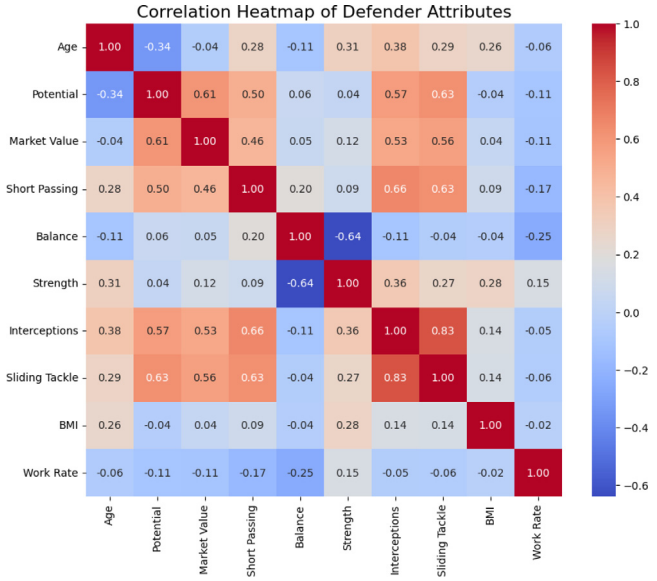


Fig. 3. Correlation matrix for selected attributes of defender.

5. Results and discussion

The target variable market value was found to be right-skewed initially. So, a natural logarithmic transformation is applied to get a normal distribution (see Fig. 2).

Then a correlation heatmap is applied to get a visual representation of the correlation matrix, displaying the correlation coefficients between multiple variables in a dataset. Each cell in the heatmap shows the correlation between two variables, with the color intensity representing the strength and direction of the correlation.

The correlation heatmaps of Figs. 3 to 6 reveal important relationships between various player attributes and market value across different positions in football.

For defenders, market value correlates strongly with potential (0.61), indicating that projected future ability significantly impacts valuation. Technical defensive skills like interceptions (0.53) and sliding tackle (0.56) show strong correlations, highlighting their

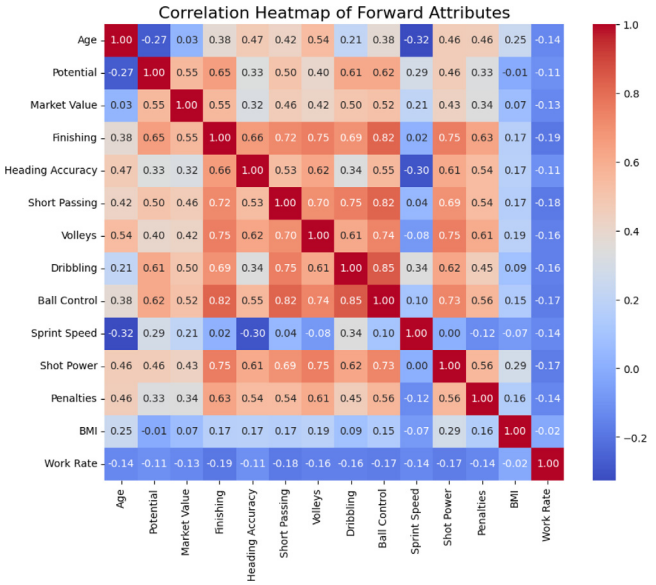


Fig. 4. Correlation matrix for selected attributes of forward.

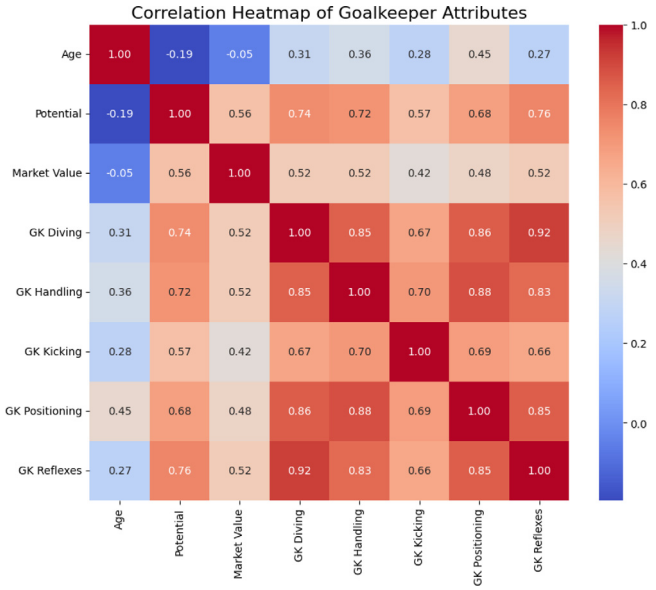


Fig. 5. Correlation matrix for selected attributes of Goalkeeper.

importance for defenders' market worth. Short passing (0.46) demonstrates a moderate correlation, suggesting that ball distribution capability adds significant value. Physical attributes like strength (0.12) show weaker correlations, while age (−0.04) has a slight negative relationship, indicating that younger defenders with development potential are valued slightly higher than older, more experienced defenders.

Forward, market value strongly correlates with potential (0.55) and finishing ability (0.55), confirming the premium placed on goal-scoring efficiency. Technical attributes like ball control (0.52), dribbling (0.50), and short passing (0.46) all show strong correlations, emphasizing the importance of technical proficiency. Physical attributes such as sprint speed (0.21) show a more moderate correlation, while age (−0.03) has a minimal negative impact on market value. However, the wide range of significantly correlated attributes suggests that forwards are evaluated on a broader skill set than other positions.

For goalkeepers, specialized skills directly correlate with market value. GK diving (0.52), GK handling (0.52), and GK reflexes (0.52) show equally strong correlations, followed by GK positioning (0.48) and GK kicking (0.42). Potential (0.56) remains a strong

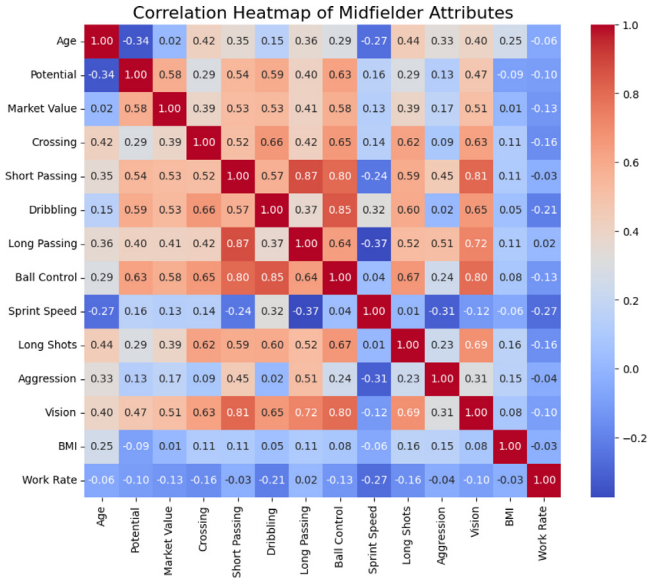


Fig. 6. Correlation matrix for selected attributes of mid-fielder.

Table 4
Results of JB Test, DW Test, t-Test, Mean and Standard Deviation for Goalkeepers.

Feature	JB Test (*p*-value)	DW test	t-Test (*p*-value)	Mean	Standard Dev
Age	35.957 (<0.01)	0.048	177.119 (<0.01)	28.05	5.33
Potential	14.995 (<0.01)	0.009	407.85 (<0.01)	72.18	5.96
GK Diving	9.0176 (<0.01)	0.009	353.672 (<0.01)	69.05	6.57
GK Handling	25.002 (<0.01)	0.009	351.253 (<0.01)	66.59	6.38
GK Kicking	39.392 (<0.01)	0.011	319.078 (<0.01)	65.09	6.86
GK Positioning	10.634 (<0.01)	0.011	316.516 (<0.01)	67.69	7.20
GK Reflexes	6.479 (<0.05)	0.010	340.044 (<0.01)	70.19	6.95

indicator of market value, consistent with other positions. The strong correlation between specific goalkeeper skills indicates that technical proficiency in position-specific attributes is highly valued in the transfer market.

The midfielders' market value correlates strongly with potential (0.58), ball control (0.58), short passing (0.53), dribbling (0.53), and vision (0.51). These correlations highlight the importance of technical skills and game intelligence for midfielders. Distribution skills like long passing (0.41) and crossing (0.39) show moderate correlations. Physical attributes like sprint speed (0.13) have weaker correlations, suggesting that technical abilities are more valued than physical attributes for midfielders.

Across all positions, potential consistently shows one of the strongest correlations with market value (ranging from 0.55 to 0.61), confirming that future development prospects significantly impact player valuation regardless of position. Age generally has a negative correlation with market value, though this effect varies by position. The work rate shows a consistent negative correlation across positions, suggesting it might be undervalued in the market. Position-specific technical skills consistently correlate more strongly with market value than general physical attributes like BMI, indicating that specialized skills drive player valuations more than physical metrics.

Table 4 summarizes statistical tests and descriptive statistics for goalkeeper attributes. The JB Test shows that none of the variables follow a normal distribution (p-values < 0.01). The DW Test indicates strong positive autocorrelation with low values (0.009–0.048). The t-test confirms significant differences for all features (p-values < 0.01). Descriptive statistics reveal that goalkeepers have an average age of 28.05 years with minimal variation (SD: 5.33). Skill attributes like GK Diving (mean: 69.05), GK Handling (mean: 66.59), GK Kicking (mean: 65.09), GK Positioning (mean: 67.69), and GK Reflexes (mean: 70.19) show moderate to high proficiency with standard deviations ranging from 6.38 to 7.20.

Table 5 provides statistical insights into defender attributes. The JB Test indicates that none of the features follow a normal distribution (p-values < 0.01). The DW Test values are low (0.006–0.061), showing strong positive autocorrelation, except for "Work Rate" (0.366). The t-test confirms significant differences for all attributes (p-values < 0.01). Defenders have an average age of 26.54 years (SD: 4.55) and a potential of 72.36 (SD: 5.41). Defensive skills like Strength (mean: 72.68, SD: 10.08), Interceptions (mean: 66.43, SD: 7.09), and Sliding Tackle (mean: 68.62, SD: 6.41) show high proficiency with moderate variation. Technical skills like Short Passing (mean: 63.12, SD: 8.14) and Balance (mean: 61.71, SD: 12.81) exhibit more variability. Work Rate averages at 5.39 (SD: 2.74), while BMI is consistent at 22.90 (SD: 1.36). Overall, defenders excel in defensive capabilities with moderate variability in technical and physical traits, though work rate and balance show greater disparities.

Table 5
Results of JB Test, DW Test, t-Test, Mean and Standard Deviation for Defenders.

Feature	JB Test	DW test	t-Test	Mean	Standard Dev
Age	134.249 (<0.01)	0.0514	380.844 (<0.01)	26.54	4.55
Potential	60.859 (<0.01)	0.009	873.387 (<0.01)	72.36	5.41
Short Passing	451.716 (<0.01)	0.013	507.123 (<0.01)	63.12	8.14
Balance	141.379 (<0.01)	0.061	315.115 (<0.01)	61.71	12.81
Strength	287.958 (<0.01)	0.037	471.346 (<0.01)	72.68	10.08
Interceptions	24.541 (<0.01)	0.016	612.778 (<0.01)	66.43	7.09
Sliding Tackle	54.360 (<0.01)	0.0130	681.208 (<0.01)	66.82	6.41
Work Rate	474.225 (<0.01)	0.366	128.549 (<0.01)	5.39	2.74
BMI	41.486 (<0.01)	0.006	1100.526 (<0.01)	22.90	1.36

Table 6
Results of JB Test, DW Test, t-Test, Mean and Standard Deviation for Mid-fielders.

Feature	JB Test	DW test	t-Test	Mean	Standard Dev
Age	204.882(<0.01)	0.044	417.0453 (<0.01)	25.63	4.56
Potential	39.466 (<0.01)	0.008	946.748 (<0.01)	73.13	5.73
Crossing	95.407 (<0.0.1)	0.027	485.369 (<0.01)	62.12	9.50
Short Passing	8.394 (<0.05)	0.007	717.169 (<0.01)	68.83	7.12
Dribbling	28.978 (<0.01)	0.013	699.278 (<0.01)	69.04	7.33
Long Passing	115.271 (<0.01)	0.017	526.914 (<0.01)	64.27	9.05
Ball Control	55.572 (<0.01)	0.007	771.423 (<0.01)	69.82	6.71
Sprint Speed	447.715 (<0.01)	0.051	446.820 (<0.01)	70.00	11.63
Long Shot	95.584 (<0.01)	0.0268	433.409 (<0.01)	61.53	10.54
Aggression	143.582 (<0.01)	0.071	307.013 (<0.01)	59.01	14.27
Vision	0.171 (<0.1)	0.011	605.874 (<0.01)	65.96	8.08
BMI	71.978 (<0.01)	0.006	1248.672 (<0.01)	22.71	1.35

Table 7
Results of JB Test, DW Test, t-Test, Mean and Standard Deviation for Forwards.

Feature	JB Test	DW test	t-Test	Mean	Standard Dev
Age	67.287 (<0.01)	0.046	242.366 (<0.01)	26.41	4.86
Potential	46.656 (<0.01)	0.008	572.643 (<0.01)	72.02	5.61
Finishing	49.282 (<0.01)	0.009	445.02 (<0.01)	69.26	6.94
Heading Accuracy	1.753 (<0.05)	0.022	344.371 (<0.01)	66.89	8.66
Short Passing	0.670 (<0.1)	0.010	352.729 (<0.01)	62.09	7.85
Volley	10.998 (<0.01)	0.020	287.034 (<0.01)	62.58	9.72
Dribbling	20.598 (<0.01)	0.009	409.261 (<0.01)	66.03	7.19
Ball Control	15.787 (<0.01)	0.007	439.216 (<0.01)	67.38	6.84
Sprint Speed	349.317 (<0.01)	0.046	277.960 (<0.01)	70.62	11.33
Shot Power	8.482 (<0.05)	0.009	414.454 (<0.01)	70.45	7.58
Penalties	23.156 (<0.01)	0.021	351.660 (<0.01)	65.52	8.31
BMI	396.117 (<0.01)	0.007	695.845 (<0.01)	23.10	1.48
Work Rate	251.512 (<0.01)	0.536	68.421 (<0.01)	4.09	2.67

Table 6 provides statistical insights into midfielder attributes. The JB Test shows that most attributes do not follow a normal distribution (p-values < 0.01). The DW Test values are low (0.006–0.071), indicating strong positive autocorrelation. The t-test confirms significant differences for all features (p-values < 0.01). Midfielders have an average age of 25.63 years (SD: 4.56) and a potential of 73.13 (SD: 5.73). Technical skills like Short Passing (mean: 68.83), Dribbling (mean: 69.04), Long Passing (mean: 64.27), and Ball Control (mean: 69.82) show high proficiency with moderate variability. Physical traits such as Sprint Speed (mean: 70.00, SD: 11.63) and Aggression (mean: 59.01, SD: 14.27) exhibit greater variability, while BMI is consistent at a mean of 22.71 (SD: 1.35). In summary, midfielders excel in technical skills with moderate variation but show notable disparities in physical attributes like Sprint Speed and Aggression, requiring advanced methods for further analysis or modeling.

Table 7 provides statistical insights into forwards' attributes. The JB Test shows that most features are not normally distributed (p-values < 0.01), except for "Short Passing" ($p > 0.1$) and "Heading Accuracy" ($p < 0.05$). The DW Test values are low (0.007–0.046), indicating strong positive autocorrelation, except for "Work Rate" (DW: 0.536). The t-test confirms significant differences across all attributes (p-values < 0.01). Forwards have an average age of 26.41 years (SD: 4.86) and a potential of 72.02 (SD: 5.61). Key attacking skills like Finishing (mean: 69.26), Heading Accuracy (mean: 66.89), and Shot Power (mean: 70.45) show high proficiency with moderate variability. Technical skills like Dribbling (mean: 66.03) and Ball Control (mean: 67.38) also exhibit strong performance, while Sprint Speed averages 70.26 with higher variability (SD: 11.33). Work Rate averages 4.09 with notable variation (SD: 2.67), and BMI is consistent at a mean of 23.10 (SD: 1.48). Overall, forwards excel in attacking and technical skills but show greater variability in physical traits like Sprint Speed and Work Rate, requiring advanced methods for further analysis or modeling efforts.

Table 8
VIF Values for All the Independent Attributes of All Categories.

Goalkeeper		Defender		Mid-fielder		Forward	
Feature	VIF	Feature	VIF	Feature	VIF	Feature	VIF
Age	3.359	Age	3.183	Age	3.191	Age	4.134
Potential	6.387	Potential	4.501	Potential	4.124	Potential	5.159
GK Diving	8.109	Short Passing	2.246	Crossing	2.435	Finishing	5.680
GK Handling	8.109	Balance	2.015	Short Passing	8.461	Heading Accuracy	2.475
GK Kicking	2.116	Strength	2.229	Dribbling	5.965	Short Passing	3.655
GK Positioning	7.709	Interceptions	4.814	Long Passing	5.419	Volley	3.515
GK Reflexes	7.779	Sliding Tackle	4.032	Ball Control	8.917	Dribbling	5.170
-		Work Rate	1.116	Sprint Speed	1.875	Ball Control	7.228
		BMI	1.166	Long Shot	1.875	Sprint Speed	1.719
				Aggression	2.396	Shot Power	3.295
-				Vision	4.311	Penalties	1.930
				BMI	1.086	BMI	1.132
				-		Work Rate	1.082

Also, the elevated VIF values suggest that the standard errors of the coefficients are magnified, resulting in broader confidence ranges. Addressing multicollinearity is crucial for enhancing the interpretability and stability of the regression model. To be more precise, when the VIF is less than 2, it indicates low multicollinearity, meaning that the variables are mainly independent of each other. When the VIF is between 2 and 5, it indicates moderate multicollinearity, which means some association among the variables. If the VIF is more than or equal to 5, it indicates a high level of multicollinearity, meaning that the variables are highly associated with each other.

Table 8 displays the VIF values for the attributes within the player categories that have been chosen, specifically Goalkeeper, Defender, Mid-fielder, and Forward. The VIF values observed in Table 3 demonstrate different levels of multicollinearity. Typically, the VIF values range from 2 to 5, which suggests the presence of substantial multicollinearity. This implies that there is a certain degree of connection between factors within these player groups. Some cases may have VIF values below 2, indicating a low level of multicollinearity. In this context, the variables exhibit a high degree of independence from one another, which helps to mitigate the potential problems associated with collinearity. Certain attributes display VIF values that surpass 5, indicating a significant presence of multicollinearity. When variables are highly correlated in such instances, it might potentially cause instability in the estimations of regression models.

Based on all insights, various ML models: MLR, RR, SVR, and RFR are applied to all categories: goalkeeper, defender, forward, and mid-fielder of football players.

Fig. 7 compares goalkeepers' predicted and actual market values in train and test sets utilizing four machine-learning models: MLR, RR, SVR, and RFR. The red dashed line indicates the best-fit line, where predictions correspond perfectly with actual values. MLR demonstrates reasonable alignment but reveals significant dispersion, especially at the extremes, highlighting its limitations for identifying complex patterns. RR enhances performance marginally by reducing this dispersion via regularization. SVR exhibits a more compact clustering of points around the best-fit line, effectively capturing non-linear relationships. The RFR exhibits superior performance, with most points closely aligned to the best-fitting line and minimum variance, underscoring its capacity to simulate intricate patterns and interactions. Non-linear models such as RFR and SVR outperform linear methods in forecasting goalkeeper market values.

Fig. 8 compares predicted and actual market values in train and test sets for defenders utilizing all models. LR and RR demonstrate a moderate correlation with a perfect fit line, although it displays significant variance, especially at extreme values. SVR shows more compact clustering around the fit line, effectively capturing non-linear correlations. The RFR has superior performance, characterized by low variance and predictions that closely align with actual values, underscoring its capacity to manage complex patterns and relationships within the data.

Fig. 9 compares predicted and actual log-transformed market values in train and test sets for midfielders utilizing all models. The performance patterns for midfielders correspond with those observed for defenders. In both cases, the RFR surpasses other models due to its capacity to manage complex patterns and relationships within the data. SVR shows robust efficiency in modeling non-linear relationships for both orientations. However, linear models such as Linear and Ridge Regression are weak in capturing non-linearities in both datasets, rendering them less useful for predicting player market values. The consistency across positions highlights the significance of non-linear models for accurate player value.

Fig. 10 compares predicted and actual market values in train and test sets for forwards using all models. LR shows moderate alignment with the perfect fit line but exhibits noticeable dispersion, especially at higher values, indicating a limited ability to capture complex patterns. RR slightly improves upon this by reducing dispersion due to regularization but still struggles with non-linear relationships. SVR demonstrates tighter clustering around the perfect fit line, effectively capturing non-linearities in the data. RFR performs the best, with minimal dispersion and predictions closely matching actual values, highlighting its strength in modeling complex interactions and patterns.

Like the defenders' and midfielders' market value predictions analyzed earlier, the forwards' market value predictions benefit significantly from non-linear models like SVR and RFR. RFR consistently outperforms other models due to its ability to handle non-linearities and feature interactions, making it the most accurate model across all positions.

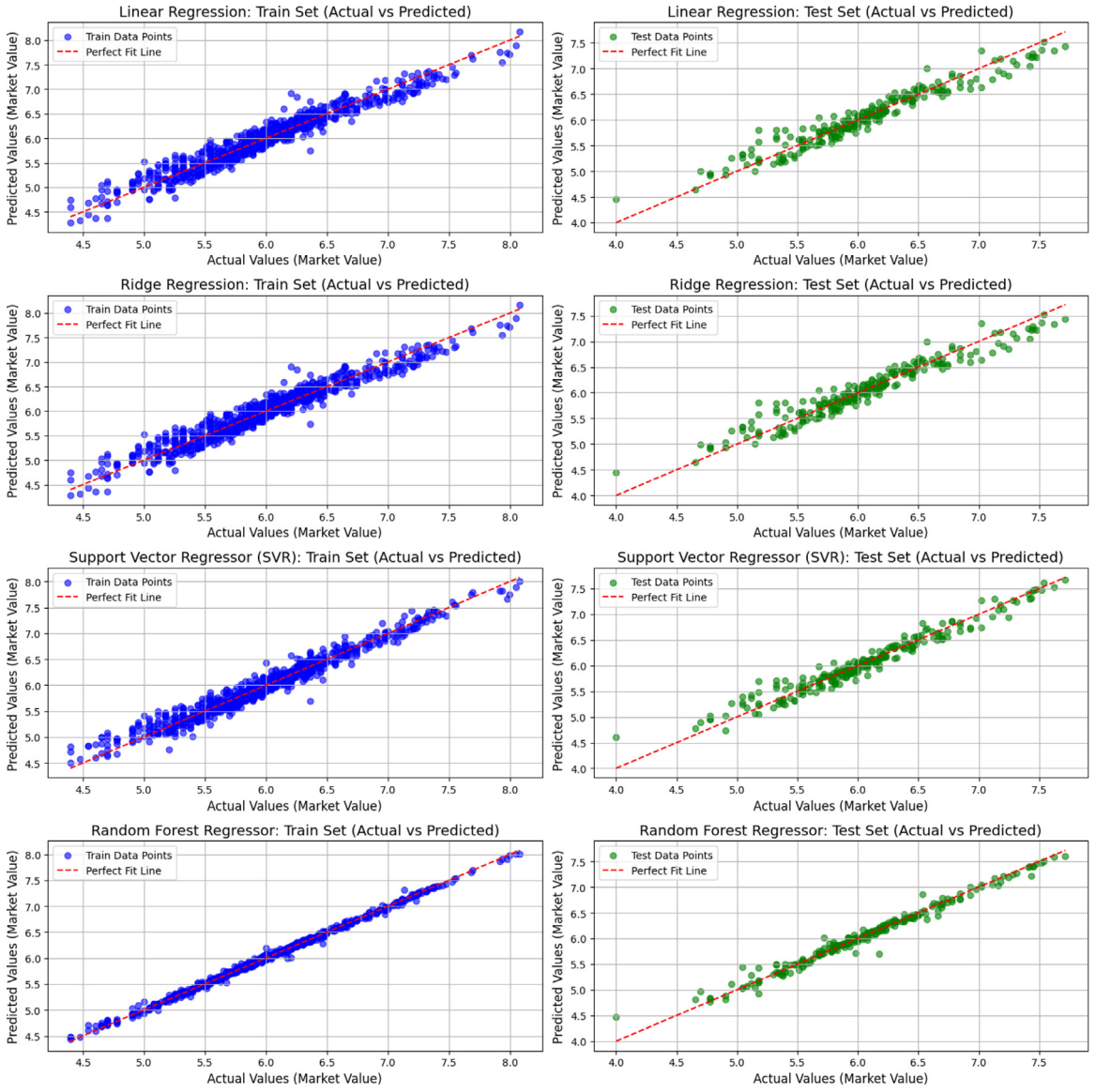


Fig. 7. Fitted curve for all models of Goalkeeper.

Table 9 provides a comprehensive comparison of evaluation metrics for four machine-learning models applied to all football player categories. The metrics include MAE, MSE, RMSE, and R^2 (across Train, Test, and Overall datasets).

For goalkeepers, the RFR achieves the best performance across all datasets. It has the lowest error values (Train MAE: 0.02, Train MSE: 0.001, Train RMSE: 0.03, Train R^2: 0.99) and maintains high accuracy in the test set (Test MAE: 0.05, Test MSE: 0.008, Test RMSE: 0.09, Test R^2: 0.97). The overall dataset results are consistent with Train and Test metrics, indicating robust generalization capability (Overall MAE: 0.02, Overall MSE: 0.002, Overall RMSE: 0.05, Overall R^2: 0.99). Other models like SVR perform well with an R^2 score of 0.95 but with slightly higher error metrics compared to RFR. MLR and RR show similar performance with moderate errors and an R^2 score of 0.93.

For defenders, RFR again demonstrates superior performance with minimal errors (Train MAE: 0.02, Train MSE: 0.001, Train RMSE: 0.03, Train R^2: 0.99) and high accuracy in the test set (Test MAE: 0.06, Test MSE: 0.008, Test RMSE: 0.09, Test R^2: 0.96). The overall dataset results align closely with train and test metrics (Overall MAE: 0.03, Overall MSE: 0.002, Overall RMSE: 0.05, Overall R^2: 0.98). SVR provides competitive results with an R^2 score of 0.93 but slightly higher error values compared to RFR across all datasets. MLR and RR lag with lower R^2 scores (0.87–0.88) and higher error values.

For midfielders, RFR consistently delivers the best results across all datasets (Train MAE: 0.02, Train MSE: 0.001, Train RMSE: 0.03, Train R^2: 0.99; Test MAE: 0.06, Test MSE: 0.009, Test RMSE: 0.09, Test R^2: 0.96; Overall MAE: 0.03, Overall MSE: 0.002, Overall RMSE: 0.05, Overall R^2: 0.98). SVR performs well with an R^2 score of 0.94 but slightly higher error values compared to

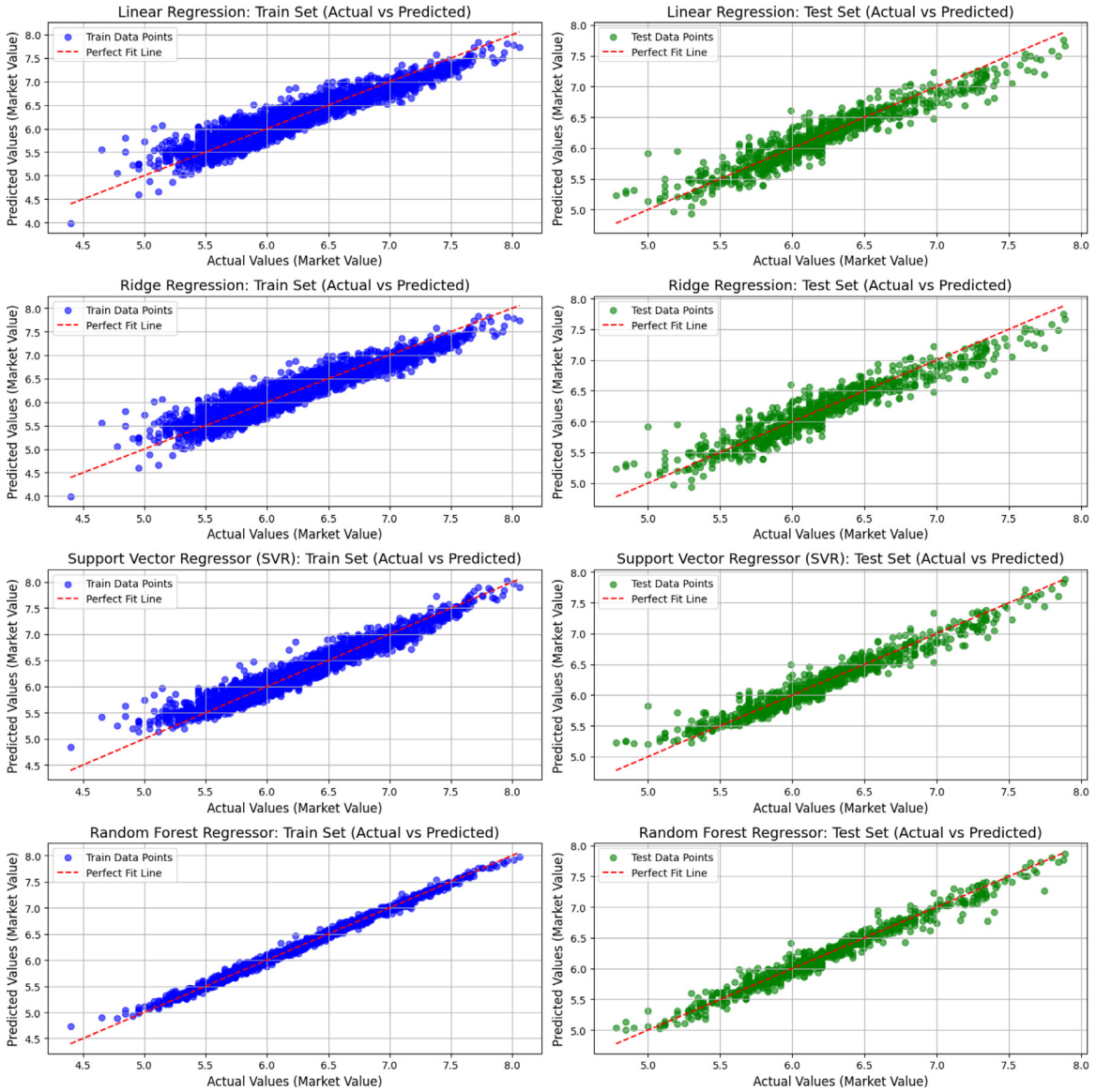


Fig. 8. Fitted curve for all models of defender.

RFR across all datasets (Train MAE: 0.08; Test MAE: 0.09; Overall MAE: 0.08). MLR and RR exhibit identical performance with an R^2 score of 0.89 but higher error metrics.

For forwards, RFR dominates across all datasets with outstanding performance metrics (Train MAE: 0.02; Train MSE: 0.001; Train RMSE: 0.03; Train R^2: 0.99; Test MAE: 0.05; Test MSE: 0.006; Test RMSE: 0.07; Test R^2: 0.97; Overall MAE: 0.02; Overall MSE: 0.002; Overall RMSE: 0.04; Overall R^2: 0.99). SVR performs well with an R^2 score of 0.96 across all datasets but has slightly higher error values compared to the RFR. MLR and RR show similar performance with an R^2 score of 0.94 but have higher error metrics compared to SVR and RFR.

Across all categories, the RFR consistently outperforms other models in terms of accuracy (highest R^2 scores) and error minimization (lowest MAE, MSE, RMSE). SVR is the second-best model in most cases due to its high accuracy but slightly higher error values compared to RFR. MLR and RR exhibit similar performance across all categories but are less accurate than ensemble-based methods like RFR or kernel-based methods like SVR.

So, it is observed that the RFR emerges as the best-performing model due to its superior accuracy and minimal errors.

Table 10 compares the performance of some existing studies and the current study in predicting player market values using various models, evaluated through R^2 and the number of features. Existing studies show varying performance, with one study using PSO and SVR achieving an R^2 of 0.740 with 55 features, while another study using Linear Regression as a baseline achieves an R^2 of 0.470 with 7 features. A third study employs Multi-level Regression but does not report an R^2 value.

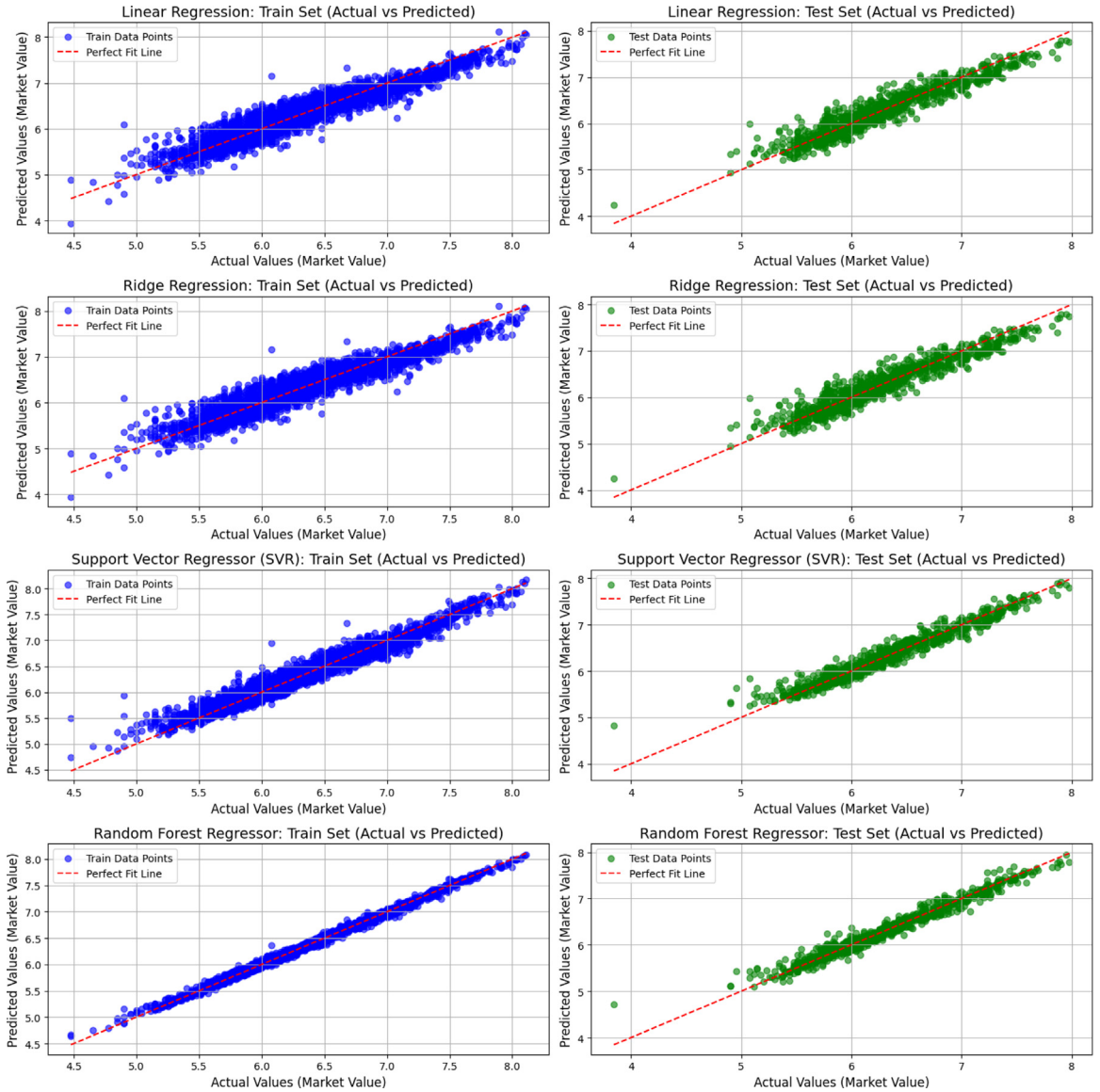


Fig. 9. Fitted curve for all models of mid-fielder.

The current study evaluates MLR, RR, SVR, and RFR across four player categories: goalkeepers, defenders, midfielders, and forwards. For goalkeepers, defenders, midfielders, and forwards, MLR and RR deliver moderate accuracy with R^2 values ranging from 0.88 to 0.94. SVR improves predictive performance with R^2 values between 0.93 and 0.96 across all categories. RFR consistently achieves the highest accuracy, with R^2 values of 0.97 for goalkeepers and forwards and 0.96 for defenders and midfielders, all while using fewer features than existing studies.

The results demonstrate that the current study outperforms previous approaches by achieving higher predictive accuracy with fewer input features. RFR emerges as the most effective model due to its ability to handle complex patterns, making it highly suitable for predicting player market values across all positions.

The results demonstrate that the current study outperforms previous approaches by achieving higher predictive accuracy with fewer input features. RFR emerges as the most effective model due to its ability to handle complex patterns, making it highly suitable for predicting player market values across all positions.

The accuracy of 0.98 was achieved by RFR for forwards, and this approach demonstrated similar high accuracies for other positions, suggesting its resilience. In addition, the study emphasized the significant factors that affect market values, such as physical, technical, and mental characteristics. In football, these insights can aid in strategic player acquisition and team optimization. Beyond sports, such predictive frameworks are widely applicable in engineering fields. For instance, RFR models are used in predictive maintenance to forecast mechanical failures in manufacturing, aerodynamic design optimization to reduce

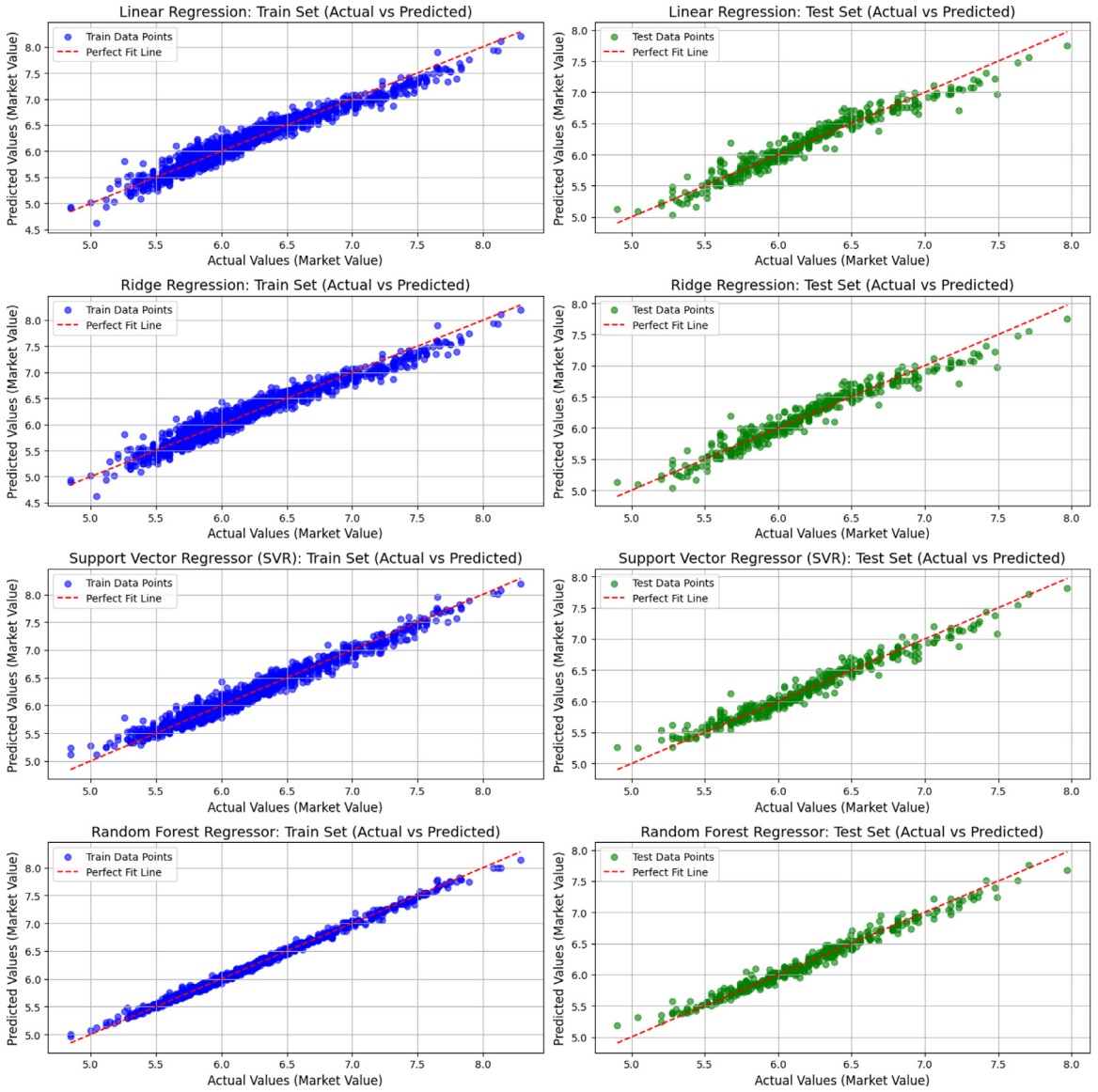


Fig. 10. Fitted curve for all models of forward.

drag in automotive components, and construction project valuation by analyzing economic indices and material costs. The universal methodology of feature importance analysis, whether evaluating a footballer's composure under pressure or identifying vibration patterns in machinery, enables data-driven decision-making across industries. These approaches not only enhance performance evaluation in sports but also drive innovation in engineering by quantifying previously intangible factors, bridging the gap between athletic analytics and industrial advancements. Also, Boosting ML models (AdaB, GBR, LGB, XGB) combined with Bayesian hyperparameter optimization accurately predict macroscopic mechanical properties of fibrous composites via RVE simulations, achieving R^2 = 0.94 to 0.99 (XGB highest) and computational efficiency over traditional methods 20 [24],21 [25],Munam et al. [28],Tariq et al. [31],25 [49],Rahman et al. [56,57],Hossain et al. [58,59],Kamruzzaman et al. [60]. Also, an Artificial Neural Network (ANN) based study investigates the erosion resistance of highway shoulder rocks under extreme rainfall using a large-scale erosion testing device (UNLETB) and trains an ANN to classify erosion characteristics based on gradation parameters (D10, D30, D60), achieving nearly 99% prediction accuracy [61]. So, boosting ML models and ANN models can be employed to achieve better performance.

Finally, the methodology and findings of this research extend beyond football, offering a versatile, ML-driven approach applicable to other sports for athlete valuation, human resources for employee assessment, real estate for property valuation, talent scouting across industries, and financial forecasting for asset valuation. The ability to objectively analyze and predict value from rich datasets has significant implications for decision-making across numerous fields.

Table 9
Results of Evaluation Metrics for all Models.

Category	Model	Train				Test				Overall			
		MAE	MSE	RMSE	R^2	MAE	MSE	RMSE	R^2	MAE	MSE	RMSE	R^2
Goalkeeper	MLR	0.11	0.02	0.14	0.93	0.12	0.02	0.16	0.93	0.11	0.02	0.15	0.93
	RR	0.11	0.02	0.14	0.93	0.12	0.02	0.16	0.93	0.11	0.02	0.15	0.93
	SVR	0.09	0.14	0.12	0.95	0.10	0.01	0.13	0.95	0.09	0.01	0.12	0.95
	RFR	0.02	0.001	0.03	0.99	0.05	0.008	0.09	0.97	0.02	0.002	0.05	0.99
Defender	MLR	0.13	0.02	0.17	0.87	0.12	0.02	0.17	0.88	0.13	0.02	0.17	0.87
	RR	0.13	0.02	0.17	0.87	0.12	0.02	0.17	0.88	0.13	0.02	0.17	0.87
	SVR	0.09	0.15	0.12	0.93	0.09	0.01	0.12	0.93	0.09	0.01	012	0.93
	RFR	0.02	0.001	0.03	0.99	0.06	0.008	0.09	0.96	0.03	0.002	0.05	0.98
Midfielder	MLR	0.12	0.02	0.16	0.89	0.12	0.02	0.16	0.89	0.12	0.02	0.16	0.89
	RR	0.12	0.02	0.16	0.89	0.12	0.02	0.16	0.89	0.12	0.02	0.16	0.89
	SVR	0.08	0.01	0.11	0.94	0.09	0.01	0.12	0.94	0.08	0.01	0.11	0.94
	RFR	0.02	0.001	0.03	0.99	0.06	0.009	0.09	0.96	0.03	0.002	0.05	0.98
Forward	MLR	0.09	0.01	0.11	0.94	0.08	0.01	0.11	0.94	0.09	0.01	0.11	0.94
	RR	0.09	0.01	0.11	0.94	0.08	0.01	0.11	0.94	0.09	0.01	0.11	0.94
	SVR	0.07	0.009	0.09	0.96	0.07	0.01	0.1	0.96	0.07	0.009	0.09	0.96
	RFR	0.02	0.001	0.03	0.99	0.05	0.006	0.07	0.97	0.02	0.002	0.04	0.99

Table 10
Evaluation of the Existing studies and our study.

Study No.	–	Algorithm Used	Test R^2	No. Features
[2]	–	A combination of PSO and SVR	0.740	55
[5]	–	Multi-level Regression	–	22
[22]	–	LR (Baseline)	0.470	7
		MLR	0.610	
		RFR	0.950	
Our Study	Goalkeeper	MLR	0.93	07
		RR	0.93	
		SVR	0.95	
		RFR	0.97	
	Defender	MLR	0.88	09
		RR	0.88	
		SVR	0.93	
		RFR	0.96	
	Midfielder	MLR	0.89	12
		RR	0.89	
		SVR	0.94	
		RFR	0.96	
	Forward	MLR	0.94	13
		RR	0.94	
		SVR	0.96	
		RFR	0.97	

6. Strengths and weaknesses

The study's main strength lies in its use of a large, detailed FIFA 22 dataset and a rigorous, position-specific machine learning approach, especially Random Forest Regression, which outperforms other models in predicting football player market values. The methodology is robust, with thorough data cleaning, feature selection, and comprehensive evaluation metrics, making the findings reliable and practically useful for football clubs. However, the study's reliance on synthetic video game data (rather than real-world transfer data) may limit real-world applicability, and issues like multicollinearity and non-normal data distributions could affect some model interpretations.

7. Future work

Future research could enhance player valuation models by integrating FIFA data with real-world match statistics, injury records, and transfer histories to improve accuracy through dynamic performance tracking. Exploring advanced techniques like neural networks might uncover deeper patterns in player attributes, while analyzing performance trends over multiple seasons could predict how players evolve or decline over time. Including broader factors like club finances, league competitiveness, and player popularity

would better reflect market realities. Making models more transparent such as showing how specific skills affect value that could increase trust among clubs. Developing specialized models for each position might yield more precise valuations, and extending the framework to predict actual transfer fees which involve negotiations and contract terms could provide direct practical value for teams.

8. Conclusion

This study confirms that machine learning, and particularly RFR, provides a powerful and accurate tool for predicting football players' market values using detailed, up-to-date data from the FIFA 22 video game. By leveraging a wide range of player attributes and position-specific features, the model delivers reliable market value estimations that can support football clubs in making informed transfer and financial decisions. The research not only advances the application of data-driven techniques in sports economics but also establishes a framework that can be adapted to various other domains where objective, data-rich valuation is needed. As ML models become increasingly sophisticated and datasets more comprehensive, the potential for accurate, automated valuation in sports and beyond will continue to grow, offering valuable insights for practitioners and researchers alike.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper and confirm that no funding was received from any agency to support this research.

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